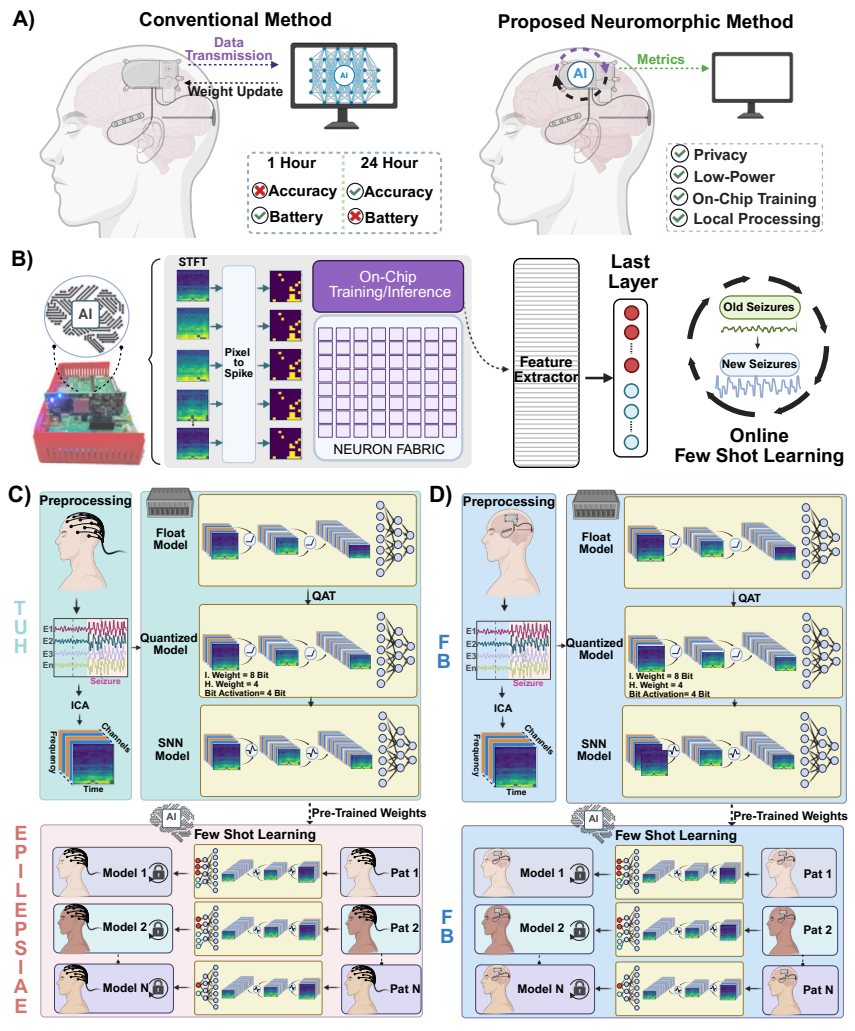
A)
Conventional Method
Data Transmission
Weight Update
AI
1 Hour
Accuracy
Battery
24 Hour
Accuracy
Battery
Proposed Neuromorphic Method
AI
Metrics
Privacy
Low-Power
On-Chip Training
Local Processing
B)
AI
STFT
Pixel to Spike
On-Chip Training/Inference
NEURON FABRIC
Feature Extractor
Last Layer
Old Seizures
New Seizures
Online Few Shot Learning
C)
TUH
Preprocessing
E1
E2
E3
En
Seizure
ICA
Frequency
Channels
Time
Float Model
QAT
Quantized Model
I. Weight = 8 Bit
H. Weight = 4
Bit Activation= 4 Bit
SNN Model
Pre-Trained Weights
AI
Few Shot Learning
EPILEPSIAE
Model 1
Model 2
Model N
Pat 1
Pat 2
Pat N
D)
FB
Preprocessing
E1
E2
E3
En
Seizure
ICA
Frequency
Channels
Time
Float Model
QAT
Quantized Model
I. Weight = 8 Bit
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Bit Activation= 4 Bit
SNN Model
Pre-Trained Weights
AI
Few Shot Learning
FB
Model 1
Model 2
Model N
Pat 1
Pat 2
Pat N